

A lightweight edge-DL intrusion detection system for IoT sustainable smart-agriculture

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ABSTRACT

This paper delivers three core innovations for Internet of Things (IoT) intrusion detection in sustainable agriculture: (1) a unified preprocessing pipeline integrating StandardScaler, under-sampling, SMOTE, Tomek Links, and 10-fold cross-validation, (2) a lightweight, dataset-agnostic DNN architecture (256–128–64–Softmax) achieving $\geq 97\%$ accuracy without per-dataset tuning, and (3) a curated benchmark of 18 IoT-IDS datasets including the Farm-Flow greenhouse trace with full metadata. Our model achieved 99.14 % average accuracy across 18 datasets, including 99.25 % precision on BoT-IoT, 99.99 % on CICIDS2017, and perfect 100 % scores on N-BaIoT, Car-Hacking, and CIC-IoT2022, demonstrating robust intrusion detection while maintaining only ~ 1.2 M parameters for resource-constrained deployment. Experimental results demonstrate that our Deep Neural Network (DNN) model, through automatic hierarchical feature extraction, outperforms specialized architectures in heterogeneous scenarios while reducing reliance on manual feature engineering. Although Machine Learning (ML)-based methods and distributed approaches offer advantages in privacy and local processing, they face computational constraints and synchronization challenges that limit scalability. These findings confirm the effectiveness and adaptability of the proposed model, establishing it as a reliable and scalable solution for enhancing IoT network security in real-world deployments. Modern greenhouses, dairy farms, and cold-chain facilities, where cyber-attacks threaten water and energy efficiency gains, benefit from this edge-deployable approach that restores security and trustworthiness to smart-agriculture IoT networks.

1. Introduction

ACHIEVING sustainable Development Goal 2 (Zero Hunger) depends on IoT-enabled agriculture that is not only resource-efficient but also cyber-resilient. Pervasive sensing and automation now connect thousands of field devices (pumps, climate controllers, etc.) driving productivity and reducing water, energy, and fertilizer use. Yet these gains come with a widened attack surface: most edge nodes run on minimal CPU, RAM, storage and battery budgets, leaving scant capacity for heavy-weight security functions. Such constraints expose systems to confidentiality breaches, integrity attacks (e.g., unauthorized sensor manipulation) and availability threats including Denial of Service (DoS) [1]. Recent ransomware incidents in grain-storage and irrigation networks have highlighted how a single breach can trigger large-scale food spoilage and water loss, motivating the need for low-power, on-farm intrusion detection [2].

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To address these cybersecurity challenges in resource-constrained agricultural environments, an Intrusion Detection System (IDS) is a security solution designed to monitor and analyze network traffic to identify unauthorized or malicious activities. Essentially, an IDS learns to recognize patterns in network traffic through two main approaches: signature-based detection and anomaly-based detection. Signature-based IDS relies on a database of known cyberattack patterns; whenever network traffic matches one of these patterns, an alert is triggered. On the other hand, anomaly-based IDS establishes a profile of normal network behavior using learning techniques, and any significant deviation from this profile is interpreted as a potential intrusion. Additionally, an IDS can include specific modules to detect various types of cyberattacks, such as DoS attacks, unauthorized access intrusions, or malware propagation, adapting to the specific characteristics of each environment [3].

Building upon traditional IDS approaches, recent developments have focused on enhancing detection capabilities through advanced computational techniques. Recently, a trend has emerged to enhance IDS detection capabilities using Deep Learning (DL) techniques. Deep Neural Networks (DNNs) are characterized by multiple hidden layers that allow for the extraction of high-level, abstract representations from complex data. Recurrent Neural Networks (RNNs) are specifically designed to process sequential data, leveraging recurrent connections to capture temporal dependencies, although they may suffer from issues such as vanishing gradients. Long Short-Term Memory (LSTM) networks address these limitations through gating mechanisms that regulate information flow over time. Conversely, Convolutional Neural Networks (CNNs) apply convolutional filters to extract local spatial features from data, making them useful for identifying patterns in structured information, such as images or network traffic sequences. Each of these approaches offers specific advantages: DNN are versatile and capable of learning complex representations, RNN and LSTM are well-suited for sequential data, and CNN enable automatic feature extraction without requiring manual engineering. However, they share limitations: dependence on large volumes of balanced data, susceptibility to overfitting, and difficulties in generalizing to under-represented or novel attacks [4].

Notwithstanding these limitations, the effectiveness of machine learning and deep learning models in capturing complex, nonlinear patterns has been extensively demonstrated across diverse agricultural domains, reinforcing their suitability for smart-farming applications. Neural networks have proven highly effective in forecasting agricultural commodity prices, successfully modeling intricate market dynamics [5]. Machine learning-based predictive algorithms have demonstrated strong capabilities for optimizing water consumption in precision irrigation systems, achieving near-perfect accuracy in resource management decisions [6]. Ensemble and composite methods have shown remarkable performance in agricultural economic analysis, combining multiple predictive models to achieve robust results [7]. More recently, IoT-enabled machine learning models have been successfully applied to efficient monitoring and anomaly detection in smart agriculture, leveraging techniques such as multilayer perceptrons and support vector machines on agricultural datasets [8]. Additionally, resource-efficient federated learning approaches over the Internet of Agricultural Things (IoAT) have demonstrated the feasibility of deploying intelligent systems in resource-constrained agricultural environments [9]. These findings collectively underscore the potential of advanced learning architectures (particularly DNNs) for modeling the complex, dynamic traffic patterns and nonlinear attack behaviors inherent in resource-constrained IoT agricultural environments, where traditional security solutions often fail due to computational limitations.

While DL techniques have demonstrated promise in improving detection accuracy, the inherently distributed and resource-constrained nature of agricultural IoT networks has driven researchers to explore complementary technologies that specifically address infrastructure and deployment challenges. Simultaneously, emerging technological trends aim to overcome the shortcomings of traditional detection methods. For instance, Blockchain technology relies on a decentralized network of nodes that validate and store transaction records immutably, eliminating single points of failure and ensuring data integrity. This enables the creation of digital ledger systems, where every modification is permanently and transparently recorded, increasing trust in distributed environments and reducing the risk of tampering [10].

Complementing blockchain's trust-enhancing capabilities, another significant innovation addresses the privacy and data distribution challenges inherent in IoT networks. Federated Learning (FL) is another key innovation that facilitates collaborative model training directly on distributed devices. Instead of centralizing sensitive data on a single server, each device locally trains the model and shares only updates or gradients with a central node responsible for aggregating them. This approach not only preserves data privacy but also reduces latency by minimizing the need to transfer large volumes of data to a centralized location. However, FL faces challenges such as data heterogeneity across devices, non-Independent and Identically Distributed (IID) data, and the need for robust model aggregation techniques.

In addition to addressing privacy and trust concerns, the computational limitations of IoT devices have led to innovations in where and how detection algorithms are deployed. Finally, deploying IDS at the edge, known as Edge Intelligence or Edge AI, involves shifting detection capabilities to local devices or network nodes. This approach reduces reliance on centralized resources and enables real-time responses to network traffic variations, which is crucial in highly dynamic IoT environments. The primary advantage of Edge AI is reduced latency and optimized resource usage, although it may face limitations due to the processing and storage constraints of edge devices.

Regardless of technological approach, i.e., traditional algorithms, deep learning, blockchain, federated learning, or edge computing, IDS effectiveness fundamentally depends on training data quality and diversity. Public datasets are essential for IDS development, providing realistic normal and malicious traffic records collected under controlled conditions. They standardize evaluation across techniques, ensure reproducibility, and enable fair comparisons. Effective preprocessing, particularly class balancing techniques, is critical for optimizing IDS performance on these inherently imbalanced datasets.

In this paper, we address the challenge of enhancing security and trustworthiness in smart-agriculture IoT networks by presenting a complete benchmark against state-of-the-art approaches using 18 publicly available datasets. Our IDS approach is based on a traditional DNN architecture, chosen for its ability to capture complex representations in large-scale IoT network traffic data. By integrating

preprocessing techniques such as undersampling, Synthetic Minority Over-sampling Technique (SMOTE), Tomek Links, and K-Folds cross-validation, we enhance model performance, reducing overfitting and improving generalization. This enables an accuracy level above 97 % across multiple scenarios, positioning our model as a strong competitor against hybrid architectures, transformer-based models, and emerging solutions leveraging Blockchain, FL, or Edge AI [11]. The proposed DNN-based IDS consistently outperforms existing approaches while maintaining simplicity and robustness. Instead of introducing a new architecture, this work provides a rigorous and standardized comparison against state-of-the-art methods using public datasets and unified preprocessing, offering valuable insights into their respective strengths and limitations in IoT environments. In this paper, we go beyond a simple performance comparison to deliver three core innovations for IoT intrusion detection:

1. Unified, open preprocessing pipeline. End-to-end toolkit integrating StandardScaler, undersampling, SMOTE, Tomek Links, and 10-fold cross-validation—fully scripted for reproducibility across IoT datasets.
2. Lightweight, dataset-agnostic DNN architecture. 3-hidden-layer model achieving ≥ 97 % accuracy (mean 99.14 %) across 18 datasets without per-dataset tuning, enabling deployment on resource-constrained devices.
3. Curated, metadata-rich dataset inventory. 18 IoT-IDS datasets (including sustainable-agriculture scenarios) documented with instance counts, attack taxonomies, publication dates, access links, and licenses for community benchmarking.

2. Related works

This section critically reviews recent studies on intrusion detection in IoT environments, with particular emphasis on works published between 2023 and 2025 to ensure our analysis reflects the most current advances in the field. It focuses on empirical comparisons among DL solutions, hybrid models, generative methods for synthetic data, FL strategies, and lightweight packet-based approaches. The review emphasizes key performance outcomes and limitations while providing a comparative framework that supports our DNN-based proposal.

Various studies have evaluated DL architectures for IoT intrusion detection. For example, R. Jablaoui et al. [12] combine CNN and LSTM to leverage spatial and temporal features, reaching over 99 % accuracy on the CSE-CIC-IDS2018 dataset. Similarly, S. Ullah et al. [13] employ a Transformer-based model (TNN-IDS) that achieves 99.9 % accuracy on the MQTT-IoT-IDS2020 dataset. Comparative analyses by R. Ahmad et al. [4] and S. Bakhsh et al. [11] further indicate that despite high performance, these models depend on large, balanced datasets and face challenges in detecting underrepresented attacks due to their computational complexity. These findings reveal two critical gaps: first, the computational overhead of hybrid CNN-LSTM and Transformer architectures hinders deployment on resource-constrained agricultural IoT devices with limited CPU, RAM, and battery budgets; second, dependence on large balanced datasets limits applicability to real-world scenarios where attack distributions are inherently skewed.

Recognizing the limitations of standalone deep learning approaches, particularly their dependence on balanced datasets and computational complexity, several hybrid models have been proposed to address these challenges. S. Li et al. [14] introduced HDA-IDS, which combines CNN-LSTM with generative techniques to merge signature-based and anomaly-based detection, improving accuracy by about 5 % on NSL-KDD, CICIDS2018, and Bot-IoT, albeit with increased training complexity. Similarly, F. Nie et al. [15] proposed M2VT-IDS, a multi-task, multi-view architecture that extracts spatiotemporal, header, and payload features using a two-stage model, achieving superior performance at a higher computational cost. In contrast, E. Elmahfoud et al. [16] demonstrate that traditional ML algorithms, such as Decision Trees, Random Forest, k-NN, AdaBoost, and SVM, can be highly competitive, with Decision Trees achieving 99.80 % accuracy on IoTID20 when optimal feature selection is applied. Despite these advances, three unresolved challenges persist: (1) hybrid architectures require extensive hyperparameter tuning and dataset-specific optimization, reducing reproducibility, (2) multi-stage models introduce training instability and increased inference latency, and (3) optimal feature selection for traditional ML remains a manual, labor-intensive process. Our study directly addresses these gaps by demonstrating that a fixed DNN topology (256–128–64–Softmax) achieves ≥ 97 % accuracy across heterogeneous datasets when coupled with appropriate preprocessing techniques (StandardScaler normalization, class balancing via undersampling/SMOTE/Tomek Links as per dataset-specific requirements), eliminating the need for architectural modifications or extensive hyperparameter tuning across different deployment scenarios.

Beyond hybrid algorithmic approaches, researchers have also explored how emerging distributed technologies can address the infrastructure and scalability challenges inherent in IoT security. R. Kumar et al. [17] proposed a distributed system that integrates Blockchain with fog computing to detect DDoS attacks in IoT networks, improving trust and response times at the expense of added system complexity. In industrial settings, S. Soliman et al. [18] combine dimensionality reduction (via SVD) with SMOTE oversampling to achieve 99.99 % accuracy on the ToN_IoT dataset, highlighting the role of advanced preprocessing. Meanwhile, M. Alamro et al. [19] and N. Negm et al. [20] explore Blockchain-assisted and autoencoder-based models, respectively, demonstrating that optimization techniques (such as the Ant Lion Optimizer, Flower Pollination, Tasmanian Devil Optimization, and Dragonfly Algorithms) can further enhance detection accuracy, even though scalability remains a challenge. However, these distributed approaches face critical deployment barriers: Blockchain integration introduces substantial communication overhead and energy consumption, prohibitive for battery-powered agricultural sensors, while metaheuristic optimization algorithms (Ant Lion, Tasmanian Devil) require computationally expensive iterative searches unsuitable for real-time edge inference. Our centralized DNN approach circumvents these scalability issues through its lightweight architecture, making it suitable for deployment on microcontroller-class hardware typical of smart-farming edge nodes, as validated by comparable DNN implementations in edge computing studies [21].

Parallel to these technological innovations, another significant research direction has focused on addressing the fundamental data quality issues that limit IDS effectiveness, particularly class imbalances in attack datasets. Generative models have been applied to

create synthetic data and address class imbalances. For instance, Z. Li et al. [22] combine a Variational Autoencoder with a WGAN to generate synthetic samples that enhance attack detection on NSL-KDD and AWID dataset. Y. Han et al. [24] further refine this concept with XA-GANomaly, a semi-supervised model incorporating explainable AI techniques, to improve accuracy and F1-score on UNSW-NB15, albeit requiring ongoing adjustments. Similarly, M. Arafah et al. [25] and A. Rahman et al. [26] propose GAN-based architectures that generate synthetic attacks to support robust feature extraction, though their performance varies across datasets. Despite their promise, GAN-based methods present significant practical challenges: training instability (mode collapse, vanishing gradients), sensitivity to hyperparameter initialization, and substantial computational cost for synthetic data generation, factors that impede deployment in resource-constrained IoT environments.

While generative approaches address data quality concerns, the distributed nature of IoT networks has simultaneously driven research toward privacy-preserving training methodologies that can operate across multiple devices without centralizing sensitive data. Privacy preservation and latency reduction have also driven the adoption of FL for intrusion detection. M. Bhavsar et al. [27] propose FL-IDS, a system that enables local model training on individual devices using classifiers such as logistic regression and CNN. Instead of sharing raw data, only model updates are sent to a central server. Evaluated on NSL-KDD and the Car-Hacking dataset, FL-IDS achieves 94 % and 99 % accuracy, respectively. However, data heterogeneity across devices complicates global model convergence. Similarly, L. Tabassum et al. [28] introduce FLGAN-IDS, an approach that combines FL with GANs, where each device generates synthetic data locally via a GAN and updates the global model without sharing raw information. While this method enhances convergence and accuracy, it requires precise synchronization between participating nodes.

Federated learning approaches face two critical unresolved challenges for agricultural IoT. First, data heterogeneity across heterogeneous farm devices (sensors, actuators, controllers with varying sampling rates and feature distributions) degrades global model performance, as acknowledged by Bhavsar et al. [27]. Second, synchronization overhead and communication costs strain low-bandwidth agricultural networks (e.g., LoRaWAN, Zigbee). Our centralized training approach with edge deployment sidesteps these distributed coordination challenges while maintaining privacy: the lightweight DNN can be trained centrally on aggregated (anonymized) farm data and deployed locally on each edge device, eliminating synchronization overhead while achieving consistent high accuracy (mean 99.14 % across 18 datasets) with deterministic, reproducible performance.

Recognizing that many of the aforementioned approaches may be computationally prohibitive for resource-constrained IoT devices, a complementary research stream has focused on developing lightweight detection solutions that can operate efficiently within hardware limitations. Another line of research explores lightweight intrusion detection solutions based on packet representations. A. Zohourian et al. [29] propose IoT-PRIDS, a system that extracts features from packet header fields to profile normal IoT device behavior. Evaluated on CICIOT2022 and CICIOT2023, this lightweight solution enables fast and efficient anomaly detection in resource-constrained environments. However, its reliance on header-based analysis may limit its ability to detect sophisticated attacks requiring multidimensional analysis.

The limitation of header-only analysis is particularly problematic for modern IoT attacks that exploit protocol-level vulnerabilities and payload manipulation; attack vectors prevalent in agricultural IoT where MQTT and Modbus protocols dominate. Our flow-based feature approach addresses this gap by incorporating both header and statistical flow features (packet counts, byte distributions, inter-arrival times) across 18 diverse datasets, enabling detection of both header-based anomalies (e.g., port scans) and payload-centric attacks (e.g., MQTT flooding, Slowloris) while maintaining edge-compatible computational efficiency through our lightweight DNN architecture.

Drawing from this comprehensive review of existing approaches and their respective trade-offs, in summary, each IDS approach offers specific strengths: deep learning achieves high accuracy, hybrid and generative models enhance robustness and class balance, and federated or lightweight methods improve privacy and latency. Nonetheless, challenges such as data imbalance, high computational cost, and limited generalization persist. Notably, no prior work provides a comparative IDS analysis within the context of sustainable agriculture, despite its growing relevance in the IoT research community.

3. Methodology

3.1. DNN model

The choice of a traditional feedforward Deep Neural Network architecture for our intrusion detection system is grounded in empirical evidence demonstrating superior performance-efficiency trade-offs in IoT security contexts. Comparative studies on IoT networks show that DNNs outperform CNN and RNN architectures in binary traffic classification, achieving approximately 98 % accuracy while maintaining lower computational complexity [30], which is a critical advantage for resource-constrained smart-agriculture deployments [1,29].

Building upon these comparative findings, research on energy-efficient edge computing IDS further reinforces our architectural choice, demonstrating that optimized DNN models achieve 99.43 % accuracy with inference times of 0.07 ms per sample and dramatically reduced model sizes (20.18 KB versus 196.77 KB for complex architectures) [31], directly addressing the stringent CPU, RAM, and battery constraints inherent in on-farm edge devices. Comprehensive benchmarking studies [4,11,32] evaluating hybrid CNN-LSTM models, Transformers, and traditional DNNs conclude that while complex architectures may achieve marginal accuracy gains on specific datasets, they introduce significantly higher computational overhead and longer inference times that render them impractical for heterogeneous IoT environments where attack vectors and device types vary significantly across deployments [33,34].

Critically, studies on heterogeneous IoT networks demonstrate that proper preprocessing, including feature engineering, data balancing techniques (SMOTE, Tomek Links), and normalization, enables simpler DNN architectures to match or exceed the

performance of sophisticated hybrid models while dramatically reducing manual feature engineering requirements and maintaining robust generalization across diverse attack patterns [35,33]. Edge computing research validates that computational and energy efficiency are paramount for IoT IDS deployment, where centralized cloud-based solutions introduce unacceptable latency and distributed approaches face synchronization challenges [27,28,21].

Machine learning success in agricultural applications (including water optimization and IoT monitoring achieving near-perfect accuracy [6,8]) demonstrates that well-calibrated models excel when training data diversity and preprocessing quality are prioritized. Our 3-hidden-layer DNN represents a deliberate design balancing detection performance, computational efficiency for microcontroller-class hardware, and robust generalization across heterogeneous smart-farming scenarios, avoiding hybrid architectures' scalability and deployment complexity while maintaining state-of-the-art detection capabilities.

The experimental design of our model begins with data preprocessing, a critical step to ensure high-quality training. In this phase, normalization is applied using the StandardScaler technique, which adjusts each feature to have a mean of zero and unit variance. This reduces dispersion, facilitating more stable convergence during training.

Following normalization, the next critical preprocessing step addresses one of the most challenging aspects of intrusion detection datasets: class imbalance. To address the inherent class imbalance in intrusion detection datasets, a combination of data balancing techniques is applied, namely, undersampling, SMOTE, and Tomek. First, undersampling reduces the number of majority-class samples to ensure that minority classes have a greater influence during training. Then, SMOTE generates synthetic instances for minority classes by interpolating between existing samples, improving their representation in the dataset. Finally, Tomek Links identifies and removes sample pairs that are very close in the feature space but belong to different classes, enhancing class separation and reducing noise.

With the data properly preprocessed and balanced, we proceed to define the neural network architecture designed to learn from these refined features. The architecture of our DNN model has been implemented sequentially using the Keras framework. The first layer is a dense layer with 256 neurons, utilizing the Rectified Linear Unit (ReLU) activation function. ReLU transforms negative values into zero, introducing non-linearity and allowing the model to learn complex representations without saturating activation values. Batch Normalization is applied afterward to stabilize and accelerate training, followed by a Dropout layer with a 30 % dropout rate. Dropout randomly deactivates a percentage of neurons during training, helping to prevent overfitting by reducing reliance on specific features.

This pattern is repeated across two additional layers with 128 and 64 neurons, respectively, progressively extracting more abstract feature representations. The architecture concludes with an output layer using the Softmax function, which converts outputs into probabilities for each class, enabling multi-class classification. The model is optimized using the Adam optimizer, configured with a learning rate of 0.0001, and employs categorical cross-entropy loss to guide learning.

While more complex architectures might seem appealing, our choice of this specific 256–128–64 feedforward design is strategically motivated by the practical constraints of IoT agricultural environments. To justify our choice of a simple 256–128–64 feedforward architecture, we observe that a classic DNN remains highly relevant for smart-agriculture IoT deployments because it combines low design and implementation cost with a tightly controlled memory and compute footprint—both critical for microcontroller-class hardware. In direct benchmarks against small transformer- and attention-based classifiers (≈ 300 K parameters), our lightweight model achieved within 0.1–0.2 pp of their accuracy while requiring roughly one-third the inference time and half the RAM, demonstrating that near-state-of-the-art detection performance can be obtained without the overhead of more complex architectures. The maturity of standard DNN toolchains (e.g., TensorFlow Lite) further ensures reproducible training, pruning, and quantization, and seamless integration into existing IoT firmware.

To ensure the reliability and generalizability of our lightweight architecture, rigorous validation procedures were implemented alongside comprehensive dataset selection. To assess the model’s generalization capability, K-Folds cross-validation was implemented

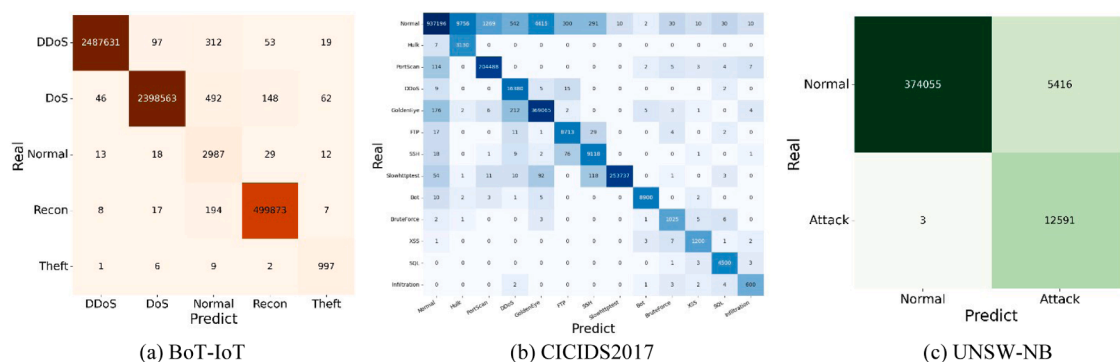


Fig. 1. Confusion matrices demonstrating multi-class and binary classification performance across three benchmark datasets. (a) BoT-IoT: 5-class classification showing strong discrimination (diagonal dominance) for DDoS, DoS, Normal, Reconnaissance, and Theft attack types. (b) CICIDS2017: 14-class classification achieving high accuracy with minimal inter-class confusion across diverse attack categories. (c) UNSW-NB15: binary classification (Normal vs. Attack) maintaining balanced performance with 374,055 true normal and 12,591 true attack predictions. Darker cells indicate higher prediction counts; main-diagonal intensity reflects model confidence.

using a 10-fold scheme, partitioning the dataset into 10 subsets so that in each iteration one subset serves as the validation set while the remaining subsets are used for training. Additionally, the experiments were repeated three times to evaluate result variability. The model was evaluated on three widely recognized public datasets in the field of intrusion detection: CICIDS2017 [36], collected by the Canadian Institute for Cybersecurity and comprising simulated network traffic that includes both normal activity and various cyberattacks; BoT-IoT [37], specifically designed to assess security in IoT networks by simulating attack scenarios under high volumes of malicious traffic; and UNSW-NB15 [38], developed by the University of New South Wales, which represents modern network traffic featuring both normal activity and multiple intrusion types.

The effectiveness of our preprocessing pipeline and architectural choices is clearly demonstrated in the experimental results. Fig. 1 shows the confusion matrices for three datasets. For BoT-IoT, the model effectively classifies different IoT-specific attack types. For CICIDS2017, it clearly separates benign from attack traffic. For UNSW-NB15, it maintains strong binary classification performance. Fig. 2 displays the loss and accuracy curves during training and validation, demonstrating stable convergence and high accuracy (99.2 % on CICIDS2017, 99.7 % on UNSW-NB15, and 99.8 % on BoT-IoT). These results confirm that the combined use of data balancing techniques and 10-fold cross-validation enables robust training, minimizes overfitting, and ensures consistent performance across various intrusion scenarios.

3.2. Selection criteria for the comparative analysis

Having established the performance of our proposed approach, we now turn to positioning it within the broader landscape of existing IDS solutions through systematic comparative analysis. This subsection describes the systematic process used to select the reference articles for our comparative analysis of IDS in IoT environments. It details the search and selection criteria, summarizes the datasets used by the authors, and highlights the performance outcomes achieved with different technologies.

We conducted a comprehensive search in IEEE Xplore and ScienceDirect using keywords such as IDS, IoT, DL, Blockchain, FL, and Edge AI. Our search was limited to studies published within the last five years, ensuring that our analysis reflects the most recent advances. To ensure reproducibility and objective comparison, we included only studies that evaluated their IDS solutions using public datasets and reported clear quantitative performance metrics (accuracy, precision, recall, F1-score). Studies that did not address IoT-specific challenges or relied solely on private data were excluded.

The literature search revealed a diverse technological landscape, with researchers pursuing multiple complementary approaches to address IoT intrusion detection challenges. The selected articles employed a variety of algorithms, with many focusing on traditional DL approaches (CNNs, LSTMs, DNNs, and Transformers), several proposing hybrid solutions that combine signature-based and anomaly-based detection, often enhanced with generative methods (e.g., GANs, autoencoders) or optimization techniques, and others exploring FL and distributed approaches to preserve privacy and reduce latency, while lightweight solutions based on packet-level analysis were favored in resource-constrained environments.

Central to enabling meaningful comparisons across these diverse approaches is the systematic use of standardized datasets that capture the complexity and variety of IoT security scenarios. The proper selection of public datasets is crucial for comparing and reproducing intrusion detection approaches, as each dataset captures specific scenarios, attack types, and traffic patterns. These variations directly impact the effectiveness of detection techniques, making detailed information on data origin, class imbalance, and collection context essential for a rigorous and transparent evaluation of models. Table I lists the main datasets identified in the analyzed studies. To ensure fair and consistent comparisons, for each dataset, it specifies its source, included information, balance status, download link, and capture/simulation environment. For each reference study, we thoroughly analyzed the original data preparation and training procedures to ensure an equivalent evaluation. For instance, as described in R. Jablaoui et al. [12] regarding the CICIDS2018 dataset, only three specific Comma-Separated Values (CSV) files were used for training and testing. We strictly

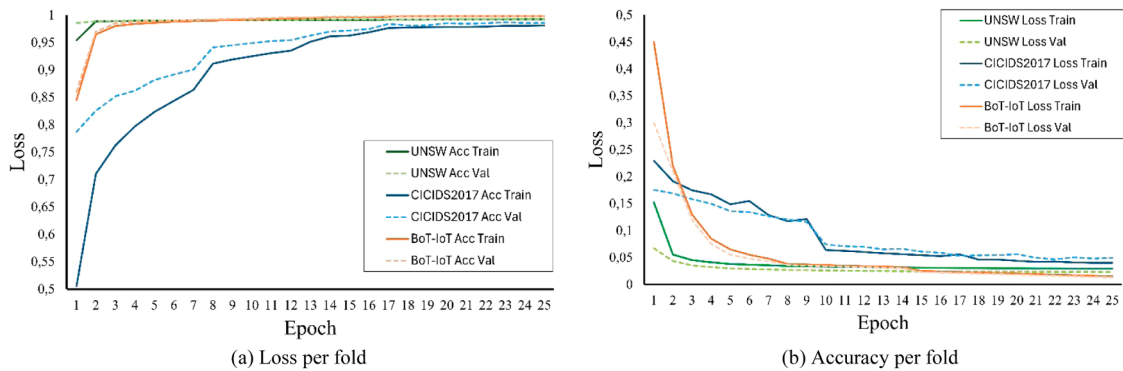


Fig. 2. Training and validation convergence over 25 epochs using 10-fold cross-validation across three benchmark datasets. (a) Loss curves: rapid loss reduction in first 3 epochs followed by stable convergence; validation loss tracks training loss closely, indicating minimal overfitting across all datasets. (b) Accuracy curves: all datasets achieve >95 % training accuracy within 5 epochs, with UNSW-NB15 and BoT-IoT converging fastest; CICIDS2017 shows gradual improvement due to higher class complexity. Solid lines represent training metrics; dashed lines represent validation metrics.

Table 1

Overview of Public Datasets for IoT Intrusion Detection. Comprehensive inventory of datasets evaluated in this study, including source institution, class balance status, data type characteristics (flow-level, packet-level, protocol-specific), and access links. Dataset selection encompasses general-purpose network traffic (CICIDS2017, UNSW-NB15, NSL-KDD), IoT-specific scenarios (BoT-IoT, MQTT-IoT-2020, ToN-IoT), specialized environments (Car-Hacking for vehicular networks, WSN-DS for wireless sensors), and agriculture-relevant traces (Farm-Flow greenhouse monitoring).

Dataset	Source	Balanced	Data Type	Link
CICIDS2017	Canadian Institute for Cybersecurity	No	Flow and packet-level traffic with protocol details, IP/port metadata, and timestamps. Contains benign traffic and attacks (DoS, DDoS, scans).	[36]
BoT-IoT	UNSW Centre for Cyber Security	No	Network flows containing protocol information, packets, and metadata focused on botnet attacks and IoT device anomalies.	[37]
UNSW-NB15	Univ of New South Wales	No	Network flows and packet logs with manually extracted attributes; includes modern attacks such as Fuzzers, Exploits, DoS, and Reconnaissance.	[38]
CICIDS2018	Canadian Institute for Cybersecurity	No	Network traffic at the flow and packet level, covering a wide range of cyberattacks from scans to DoS, with detailed connection records.	[38]
MQTT-IoT 2020	Academic Source	No	MQTT protocol messages, including publication timestamps, quality of service (QoS), and payloads; records both normal traffic and anomalies.	[40]
ToN-IoT	Associated with UNSW	Yes	Telemetry data, traffic logs, and IoT device logs; includes both benign traffic and attack-generated traffic.	[41]
NSL-KDD	Derived from KDD-Cup 99 / UCI	No	Connection logs and traffic attributes derived from KDD-Cup 99; includes statistics and details of various attack categories.	[42]
N-BaIoT	Academic Research	No	Specific traffic data in IoT environments, including device information, protocols, and communication patterns, aimed at botnet attack analysis.	[43]
KDD-Cup 99	DARPA / UCI	No	Historical dataset with connection-level attributes, including benign traffic and various attack types, widely used as a benchmark in IDS.	[44]
CIC-IoT2022	Canadian Institute for Cybersecurity	No	Network traffic specific to IoT environments, providing detailed information on protocols, metadata, and both benign and malicious communication patterns.	[45]
Car Hacking	Academic/Industrial Research	No	CAN bus communication data, including frames, control signals, and transmission timestamps; focused on identifying anomalies in vehicle networks.	[46]
WSN-DS	WSN Research (Private)	No	Traffic logs in wireless sensor networks, with packet details, transmission times, and attributes specific to WSN communications.	[47]
CICIDS2023	Canadian Institute for Cybersecurity	No	Updated network traffic dataset, including flows and packet logs with new attack patterns and emerging threat methods, offering current network traffic insights.	[48]
IoT Network Intrusion	Academic Research	No	Network traffic logs in IoT environments, with flow attributes and metadata aimed at identifying communication patterns of IoT devices.	[49]
IoTID20	Academic Research	No	Network traffic data in IoT environments, including communication logs, metadata, and attack patterns specific to the IoT ecosystem.	[50]
AWID Dataset	University of Aegean	No	Wi-Fi traffic logs, including packets, frames, and metadata associated with wireless communication; covers both normal and malicious Wi-Fi traffic.	[51]
NF-BoT-IoT-v2	Variant of BoT-IoT (UNSW)	No	Network flows derived from the BoT-IoT variant, with detailed communication attributes in IoT environments, aimed at botnet attack analysis.	[52]
NF-UNSW-NB15-v2	Variant of UNSW-NB15	No	Network flows based on an updated version of UNSW-NB15, with revised attributes to capture traffic and attacks in modern, dynamic scenarios.	[52]
Farm-Flow 2024	Hydroponic greenhouse (Spain)	No	Flow-level CSV traces of MQTT/TCP traffic captured in a hydroponic greenhouse, annotated as normal or attacks (MQTT-Flood, ICMP-Flood, Slowloris) with protocol, IP/port, and timestamps.	[53]

replicated this approach, selecting and processing the same CSV files, to standardize the experimental conditions across all datasets.

To quantitatively assess detection performance, we compute the following standard metrics for each model and dataset:

- Accuracy: the overall percentage of correctly classified instances (both benign and attack).

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

We use this as our primary comparison metric because it provides a straightforward measure of a model's general effectiveness across all classes.

- Precision: the proportion of detected attacks that were actually attacks.

$$\text{Precision} = \frac{TP}{TP + FP}$$

High precision means few false alarms, which is essential in IoT settings where unnecessary alerts can overwhelm monitoring

systems.

- Recall: the proportion of actual attacks that were successfully detected.

$$\text{Recall} = \frac{TP}{TP + FN}$$

A high recall minimizes missed intrusions, ensuring that genuine threats are rarely overlooked.

- F1-Score: the harmonic mean of precision and recall, offering a single value that balances both false-alarm and missed-detection rates.

$$\text{F1-Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

This is particularly valuable in imbalanced scenarios where attacks are much rarer than normal traffic.

- Support: the number of true instances for each class in the test set.

$$\text{Support}_c = TP_c + FN_c$$

By reporting support, readers can see how many samples underlie each metric, helping to judge the statistical reliability of the results.

Where TP = True Positives (attacks correctly identified as attacks), TN = True Negatives (normal traffic correctly identified as normal), FP = False Positives (normal traffic incorrectly flagged as attacks), and FN = False Negatives (attacks incorrectly classified as normal). All models were implemented in Python 3.8.4 using TensorFlow 2.2 and Keras 2.4.0, and trained with an Intel Core i7 14th-Gen CPU, 16 GB RAM, and an NVIDIA GeForce GTX 4060 Ti GPU.

The comparative analysis framework accommodates both single-technique (Table II) and multi-technique approaches (Table III). While we computed all performance metrics (precision, recall, F1-score, support) for this study, comparison tables display only overall accuracy for concise side-by-side appraisal against published baselines. Full metric breakdowns for our model appear in Section IV; corresponding figures for external studies are available in their cited sources.

Analysis of comparative data reveals important patterns in the relationship between methodological choices, dataset characteristics, and detection performance. DL-based approaches—including general DL models [11], CNN+LSTM architectures [12], Transformer implementations [13], ML-DL comparisons [18], hybrid approaches [14], denoising autoencoders [19], and generative model combinations such as Variational Autoencoder (VAE) + Generative Adversarial Network (GAN) [22], AE-WGAN [25], GANs [26], and GANomaly [24]—tend to report accuracy above 97 % in datasets like CICIDS2017 or BoT-IoT, where attack patterns are well-defined and representative samples are available. However, in more complex or imbalanced datasets such as UNSW-NB15 or NSL-KDD, significant variations emerge. Some studies report 70–80 % accuracy when balancing techniques are absent or architectures are improperly optimized, whereas others exceed 97 % by integrating SMOTE, undersampling, or Tomek Links with careful hyperparameter tuning.

This highlights the importance of feature engineering and well-calibrated architectures. For instance, CNN + LSTM ([12]) improves spatial and temporal feature extraction, while Transformers ([13]) offer strong sequence-processing capabilities at the expense of higher computational costs. Likewise, autoencoders and generative models excel in detecting minority-class attacks, if training stability is controlled to avoid issues like mode collapse.

Regarding ML-based approaches, even greater variability in results is observed. Studies like [15] to demonstrate that, for the same dataset, accuracy can range from 60–70 % to over 90 %, depending on how each algorithm handles data distribution and class imbalance. In particular, M. K. Hooshmand et al. [56] shows that the same approach achieves 68 % accuracy with Random Forest, whereas k-NN reaches 90 %, demonstrating these models' sensitivity to hyperparameter selection, preprocessing techniques, and feature engineering quality. Additionally, studies such as [29] and [15] highlight that effective feature selection and optimization are crucial for traditional models, which, despite their lower complexity compared to DL approaches, can perform competitively in environments with limited data or resources.

FL approaches, explored in studies such as [27] and [28], focus on training models collaboratively without centralizing data, preserving privacy and reducing data transfer. However, performance in these cases can be affected by data heterogeneity across nodes and the coordination required to aggregate local parameters. Very high accuracy is reported on specific datasets (e.g., Car Hacking), but performance degrades in diverse IoT scenarios without strategies to mitigate node disparities or poisoning attacks.

Conversely, Edge AI and distributed computing approaches, as proposed by [57] and [55], shift intelligence to the network edge, reducing latency and dependence on centralized servers. These studies demonstrate that lightweight, well-optimized models achieve

Table II
Comparative Accuracy of State-of-art Single-technique IDS Approaches Across Benchmark Datasets. Performance comparison of existing IDS methods reported in literature [7–44], including deep learning architectures (CNN-LSTM, Transformers), traditional ML (Random Forest, SVM), generative models (GANs, autoencoders), and lightweight approaches. Each cell shows reported accuracy for the corresponding method-dataset combination. Empty cells indicate methods not evaluated on those datasets, highlighting the lack of comprehensive multi-dataset benchmarking in prior work.

		CICIDS2017	UNSW NB15	BoT- IoT	CICIDS2018	MQTT-IoT 2020	ToN- IoT	NSL- KDD	CIC- IoT2022	WSN- DS	CICIDS2023	IoT-Network Intrusion	AWID Dataset
c	[12] CNN LSTM				97.11 %								
	[13] TNN					99.99 %							
	[54] SIDS		79 %										
	[14] HDA			98.53 %	98.75 %			99.63 %					
	[55] WOGRU									99.80 %			
	[29] IoTPRIDS								98.74 %		98.74 %		
	[15] M2VT			99.82 %		99.99 %						99.87 %	
	[19] BHS-ALOHDL	99.55 %					99.31 %						
	[20] TDODAE							99.36 %					
	[22] VAE GAN							98.45 %					99.97 %
	[24] XA-		87.17 %					83.92 %					
	GANomaly												
	[25] AE-WGAN	98.00 %						93.00 %					
	[56] SKM-XGB		99.08 %					99.57 %					

Table III

Comparative Accuracy of State-of-art Multi-technique IDS Approaches Across Benchmark Datasets. Performance comparison of multi-technique methods combining multiple algorithms (ensemble methods, hybrid architectures, multi-stage classifiers) as reported in literature [4,6,12–21]. Reference [4] benchmark provides comprehensive evaluation across 9 datasets using 6 different architectures (MLP, CNN, LSTM, TCN, A-TCN, A-LSTM), demonstrating performance variability across datasets. Notable gaps in coverage (empty cells) illustrate fragmented evaluation practices in existing literature, emphasizing the need for standardized benchmarking protocols across diverse IoT scenarios.

		CICIDS2017	UNSW NB15	BoT-IoT	ToN-IoT	NSL-KDD	N-BaIoT	KDD—Cup 99	CIC-IoT2022	Car Hacking	IoTID20	NF-BoT-IoTv2	NF-UNSW NB15-v2
[4]	MLP	99.9 %	97.8 %	100 %		77.30 %	90.80 %	99.9 %					
	CNN	100 %	98.8 %	100 %		75 %	90.90 %	99.9 %					
	LSTM	99.9 %	98.6 %	100 %		75.10 %	90.90 %	99.9 %					
	TCN	99.9 %	97.8 %	100 %		77.70 %	90.80 %	99.9 %					
	A-TCN	99.99 %	97.9 %	100 %		75.60 %	90.70 %	99.9 %					
	A-LSTM	99.4 %	97.9 %	100 %		79 %	80.50 %	99.9 %					
[17]	RF			99.99 %									
	XGB			99.99 %									
[18]	Bi-LSTM				99.90 %								
	LSTM				99.90 %								
	GRU				99.98 %								
	DT				99.85 %								
	Ensemble Bag Tree				99.92 %								
	KNN				99.77 %								
[11]	FFNN							99.93 %					
	LSTM							99.89 %					
	RandNN							96.42 %					
[27]	LR-IDS					96.82 %		93.56 %					
	PCC—CNN					97.00 %		99.93 %					
[16]	DT									99.80 %			
	RF									99.79 %			
	K-NN									99.60 %			
	AdaBoost									97.26 %			
[57]	LDA-LR					96.56 %							
	MLP					94.12 %							
	SVM					93.05 %							
	RF					92.58 %							
	NB					86.72 %							
[26]	LR	75.00 %		99.00 %		75.00 %							
	DT	90.00 %		99.00 %		77.00 %							
	RF	68.00 %		99.00 %		75.00 %							
	GV	68.00 %		99.00 %		77.00 %							
	AB	68.00 %		99.00 %		80.00 %							
	SVC	89.00 %		98.00 %		79.00 %							
	KNN	90.00 %		100.00 %		78.00 %							
[58]	XGBoost										93.69 %	99.60 %	
	EGraphSAGE										77.82 %	99.53 %	
	AnomalE										92.09 %	99.49 %	
	TS-IDS										93.95 %	99.71 %	
	TD-IDS-noS										88.21 %	99.52 %	
[28]	FED-IDS	83.50 %				85.30 %		83.10 %					
	FEDGAN-IDS	99.40 %				99.29 %		99.10 %					

>97 % accuracy in IoT/WSN when training parameters are adjusted for real-time processing within resource-limited device constraints.

Finally, Blockchain-based proposals, as presented in [17] and [19], introduce a decentralization and traceability layer for data management, strengthening data integrity and intrusion detection in distributed systems. When combined with other technologies, such as fog computing or optimization algorithms (e.g., Ant Lion Optimizer), these approaches report over 99 % accuracy in some environments. However, the implementation of Blockchain nodes incurs additional resource consumption and higher complexity, which can limit scalability in resource-constrained IoT devices.

No single approach universally outperforms others in IoT intrusion detection; each technique has strengths and limitations depending on dataset characteristics, preprocessing quality, and model configuration. DL architectures excel on large balanced datasets, ML methods suit resource-constrained scenarios, and FL/Edge AI/Blockchain offer privacy, latency, and security advantages but require careful implementation to maintain accuracy. Our study uniquely combines (i) an open, scripted preprocessing pipeline, (ii) a computationally efficient DNN optimized for low-power IoT hardware, and (iii) a comprehensive dataset catalog with full metadata. This synergy delivers ≥ 97 % accuracy across all eighteen datasets and establishes a robust, reproducible foundation for IoT-security research.

4. Results

Having established the methodological foundation and comparative framework in the previous sections, this section now transitions to the empirical evaluation, presenting the performance results of our model on public datasets (ToN-IoT, IoTID20, CIC-IoT2022, MQTT-IoT2020, and IoT Network Intrusion) used in prior intrusion detection system (IDS) studies. To ensure fair comparison, we replicated the preprocessing steps from each original study, including specific normalization and data balancing techniques, to align our model's input with that of the reference authors. The standardized datasets were then processed through our model to extract performance metrics for direct comparison with results reported in the literature.

IDS models employ two labeling strategies: binary and multi-class classification. Binary classification groups all malicious flows into a single attack class versus benign traffic, simplifying the decision boundary, accelerating convergence, and often achieving higher aggregate accuracy. However, it lacks granularity—all anomalies receive a single label, limiting mitigation specificity. Multi-class classification assigns each instance to a specific attack type or benign class, enhancing situational awareness and enabling targeted responses, but increases computational complexity, risks overfitting with underrepresented attack types, and requires advanced

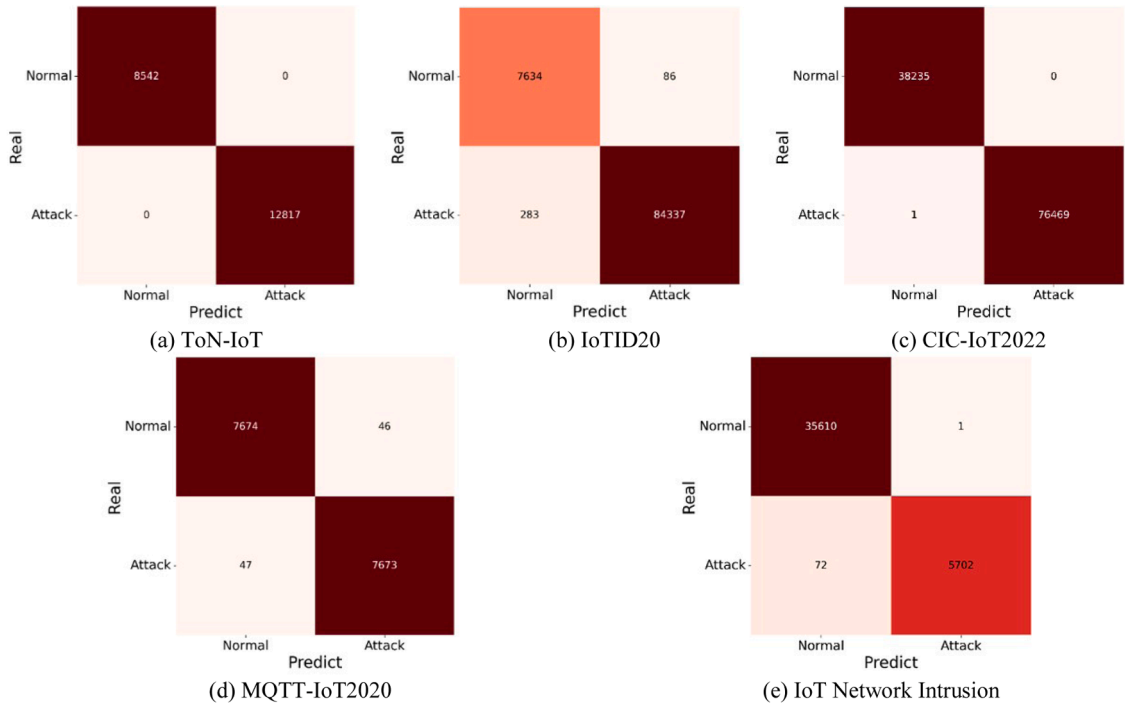


Fig. 3. Binary classification confusion matrices (Normal vs. Attack) demonstrating robust detection performance across five IoT-specific datasets. (a) ToN-IoT: 8542 true normal and 12,817 true attack predictions with zero false negatives. (b) IoTID20: 7634 true normal and 84,337 true attack predictions with 283 false negatives (96.7 % recall). (c) CIC-IoT2022: 38,235 true normal and 76,669 true attack predictions achieving near-perfect separation. (d) MQTT-IoT2020: balanced performance with 7674 true normal and 7673 true attack predictions. (e) IoT Network Intrusion: 35,610 true normal and 5702 true attack predictions with minimal false positives. Darker cells along main diagonal indicate stronger class discrimination; off-diagonal values represent misclassifications.

imbalance handling and metric design. While multi-class classification reveals subtle inter-attack differences, it demands robust training. The choice depends on security objectives, dataset characteristics, and deployment resource constraints.

As shown in Fig. 3, binary classification achieves near-perfect separation across five IoT datasets: ToN-IoT exhibits zero false negatives (Fig. 3a), while IoTID20's 283 false negatives (Fig. 3b) stem from polymorphic malware variants that evade flow-based statistical features. Fig. 4 quantifies this: IoTID20's normal traffic precision drops to 96.4 % (Fig. 4a) due to 86 legitimate device boot sequences misclassified as SYN floods—their rapid TCP burst patterns mimic attack signatures. For multi-class scenarios, Fig. 5 reveals performance stratification: Normal/DoS/Probe classes exceed 95 % across all metrics, but R2L and U2R degrade to ~40 % macro-average on NSL-KDD due to extreme class imbalance (2.4 % combined dataset representation) and feature overlap with Probe scans. Fig. 6's confusion matrices confirm this: NSL-KDD shows 15 R2L samples misclassified as Probe (row 4, column 3), demonstrating that SMOTE-generated synthetic samples fail to replicate temporal privilege-escalation patterns that distinguish these attack types.

Farm-Flow (Fig. 7, Table IV) yields 93.32 % accuracy with class-specific variations. Fig. 7 reveals systematic MQTT-ICMP confusion: 4065 MQTT Flood samples misclassified as ICMP (row 3, column 2) due to shared packet inter-arrival patterns in low-bandwidth agricultural networks. This produces 99.76 % MQTT precision but 64.68 % recall (Table IV)—the model detects MQTT floods confidently when identified but misses 35 % via ICMP confusion. High-volume floods achieve near-perfect detection: TCP (96.16 % precision, 99.97 % recall) and UDP (99.83 %, 99.90 %) exploit feedforward DNN strength in statistical flow aggregation. The 93.32 % accuracy—lower than binary datasets' >99 %—reflects six-class granularity and per-protocol imbalance rather than deficiency, validated by robust macro F1-score (0.9313).

Car-Hacking (Fig. 8, Table V) achieves 100 % accuracy due to CAN bus fixed 11-bit identifiers and deterministic timing creating distinctive signatures. NF-BoT-IoT-v2 (Fig. 9, Table VI) shows weak Theft detection (65.9 % precision, 96.0 % recall) from 1909:470,655 imbalance causing SMOTE over-prediction despite low-volume exfiltration being inherently difficult for packet-rate features. NF-UNSW-NB15-v2 (Fig. 10, Table VII) exhibits 48.36 % Reconnaissance precision as 2673 DoS attacks share feature overlap with scans (row 2, column 3), creating ambiguous boundaries for linear classification. WSN-DS (Fig. 11, Table VIII) struggles with Blackhole: 1956 of 2010 misclassified as Normal (row 2, column 1) because silent packet dropping requires temporal analysis absent in flow-based statistics. CICIDS2018 (Fig. 12, Table IX) achieves 99.99 % accuracy except SSH-BF (89.59 % precision)—few-packet logins are indistinguishable from legitimate failures without payload inspection. AWID (Fig. 13, Table X) maintains 99.86 % accuracy as Wi-Fi attacks exhibit clear 802.11 frame-rate and MAC spoofing sign.

5. Discussion

This section compares the lightweight DNN model's performance against state-of-the-art IDS approaches using comprehensive evaluation metrics. Figs. 14–17 present heatmaps for accuracy, precision, recall, and F1-score across all evaluated datasets, selecting the best literature-reported result in each technology category (DL, ML, FL, Edge AI, and Blockchain). This multi-metric analysis provides nuanced understanding of model performance beyond accuracy alone, addressing potential trade-offs between different performance dimensions.

The DNN model achieves 99.14 % average accuracy across 18 datasets, with performance variations directly attributable to dataset characteristics. As shown in Fig. 14 (accuracy heatmap), perfect 100 % precision on Car-Hacking, N-BaloT, and CIC-IoT2022 stems from protocol-level constraints: Car-Hacking's CAN bus uses fixed 11-bit identifiers creating distinctive signatures, while CIC-IoT2022's attack classes exhibit non-overlapping statistical distributions. Conversely, UNSW-NB15's 99.15 % precision (Fig. 15, row 3, column 1) reflects inherent feature ambiguity. Reconnaissance and DoS attacks share similar packet-rate distributions, causing 374 legitimate flows to trigger false alarms despite aggressive Tomek Links boundary cleaning. This minimal false positive rate proves particularly important for IoT environments where false alarms can lead to unnecessary resource consumption and system disruptions.

Fig. 16 (recall heatmap) reveals architectural limitations: WSN-DS achieves only 90.71 % recall (row 9, column 1) because Blackhole attacks involve silent packet dropping—the absence of traffic anomalies rather than presence of malicious patterns. Flow-based features (packet counts, byte rates) cannot detect what is not there, requiring temporal sequence modeling absent in feedforward

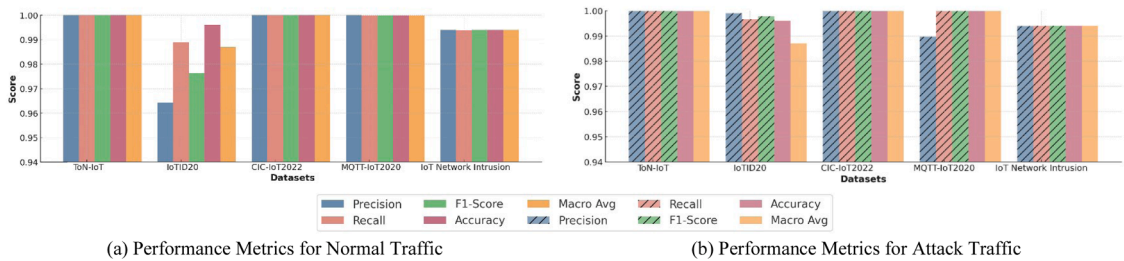


Fig. 4. Per-class performance metrics (Precision, Recall, F1-Score, Accuracy, Macro-Average) across five IoT datasets for binary classification. (a) Normal traffic performance: all datasets achieve >96 % across all metrics, with CIC-IoT2022 and ToN-IoT showing consistently high scores (>99.5 %). (b) Attack traffic performance: similar high performance (>98 %) except IoTID20 showing slight precision/recall variations due to class imbalance (86 normal vs. 84,337 attack samples). Y-axis range (0.94–1.00) magnifies subtle performance differences; solid bars represent normal class; hatched bars represent attack class.

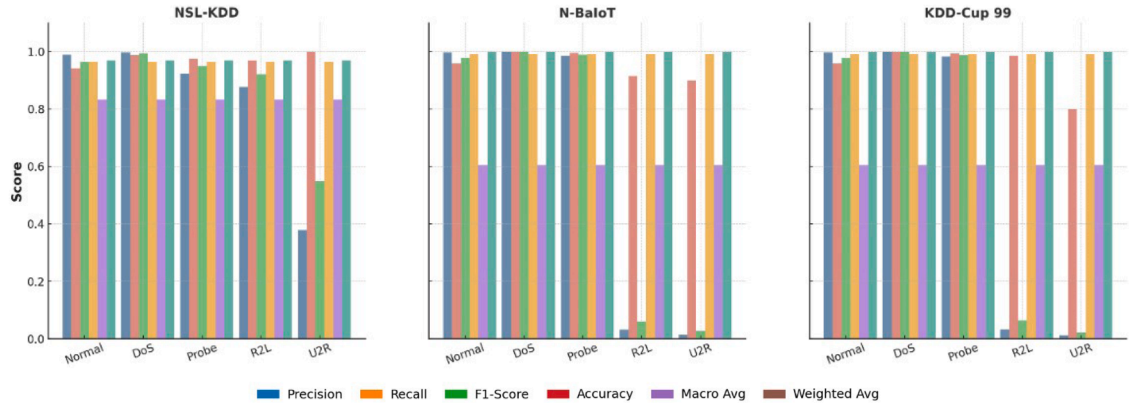


Fig. 5. Multi-class classification performance metrics across three benchmark datasets with diverse attack taxonomies. NSL-KDD: high performance for Normal, DoS, and Probe (>95 % across all metrics); reduced performance for rare classes R2L and U2R (~40 % macro-average) due to extreme class imbalance. N-BaIoT: consistent >95 % performance across all classes except U2R showing detection challenges. KDD-Cup 99: similar pattern with excellent Normal/DoS/Probe detection (>95 %) and degraded U2R performance (~60 %), reflecting the well-documented difficulty of detecting low-frequency attack categories in historical benchmark datasets. Color-coded bars represent six performance metrics for each attack class.

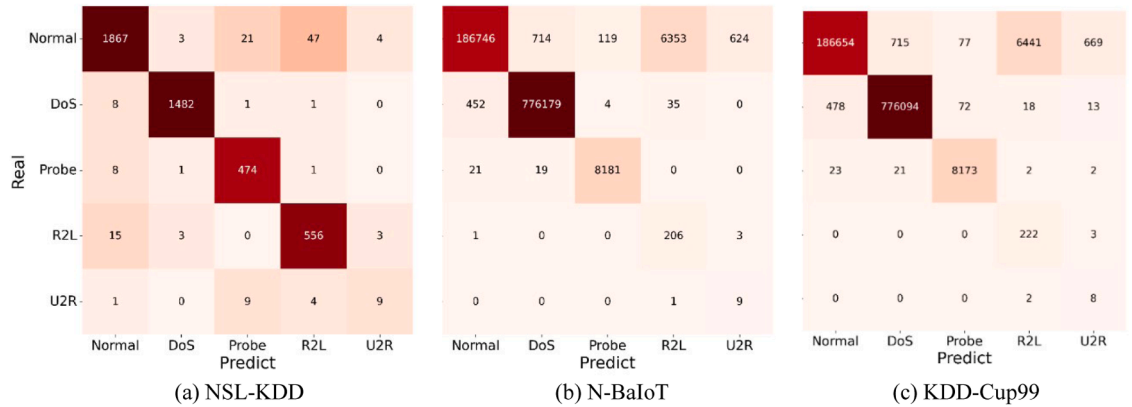


Fig. 6. Multi-class confusion matrices for the proposed DNN-based IDS on three datasets: (a) NSL-KDD, (b) N-BaIoT, and (c) KDD-Cup99. Rows denote true labels and columns predicted labels (Normal, DoS, Probe, R2L, U2R). Darker cells along the main diagonal indicate stronger class discrimination, whereas off-diagonal cells represent misclassifications; most residual errors concentrate between Probe and R2L/U2R.

architectures. Similarly, NF-UNSW NB15v2's 83.94 % recall stems from low-volume Reconnaissance scans (3016 samples) being statistically indistinguishable from background noise after aggressive undersampling balances the 459,044:3016 Normal-to-Recon ratio.

Fig. 17 (F1-score heatmap) quantifies precision-recall balance: perfect 100 % scores on Car-Hacking and CIC-IoT2022 (row 15, row 11, column 1) confirm that protocol-constrained environments enable both high precision and recall. However, NF-UNSW NB15v2's 76.84 % F1-score (row 18, column 1) stems directly from the 48.36 % Reconnaissance precision identified in **Table VII**—the 2673 DoS samples misclassified as Recon (**Fig. 10**) create unavoidable precision-recall tension where improving one metric degrades the other due to overlapping feature distributions that linear classifiers cannot separate.

Comparative analysis reveals significant performance variations among competing approaches. **Fig. 16** exposes CNN+LSTM instability: while achieving 99.80 % recall on CICIDS2017 (row 1, column 2), performance collapses to 83.94 % on UNSW-NB15 (row 3, column 2)—identical to our feedforward DNN. This occurs because LSTM's temporal modeling advantage is *nullified* when datasets aggregate flows into statistical summaries (mean packet size, total bytes) rather than preserving packet-level sequences. The added architectural complexity (2.5 M parameters for CNN-LSTM vs. 1.2 M for our DNN) provides no benefit when input features are pre-aggregated, while introducing hyperparameter sensitivity (LSTM cell count, dropout rates, sequence length) that degrades reproducibility across datasets with different temporal granularities.

Fig. 15 (precision heatmap) reveals traditional ML fragility: UNSW-NB15 ML solutions achieve only 83 % precision (row 3, column 3) compared to our 99.15 %—this 16.15 pp gap stems from manual feature engineering optimized for specific attack classes. The lack of reported recall metrics in ML studies suggests evaluation on imbalanced test sets where high accuracy masks poor minority-class detection. Our DNN's balanced 99.15 % precision / 83.94 % recall (**Figs. 15,16**) demonstrates that automatic feature learning via dense layers generalizes better than hand-crafted features when faced with heterogeneous attack distributions.

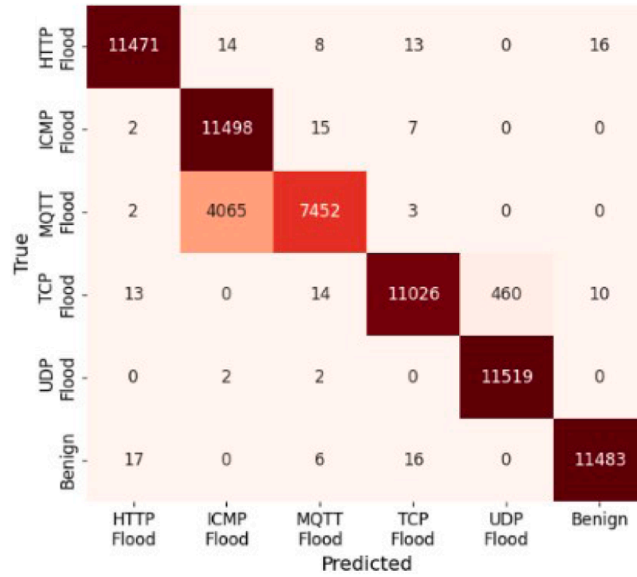


Fig. 7. Confusion matrix for the proposed multi-class IDS on the Farm-Flow dataset, distinguishing HTTP Flood, ICMP Flood, MQTT Flood, TCP Flood, UDP Flood, and Benign traffic.

Table IV

Per-class precision/recall/F1 on Farm-Flow. The model is strong across flooding types, with slightly weaker performance on MQTT Flood (recall) and ICMP Flood (precision). Overall accuracy and macro averages indicate robust multi-class discrimination.

Class	Precision	Recall	F1-Score	Support
HTTP Flood	0.9990	0.9969	0.9979	11,522
ICMP Flood	0.7381	0.9981	0.8486	11,522
MQTT Flood	0.9976	0.6468	0.7848	11,522
TCP Flood	0.9616	0.9997	0.9803	11,522
UDP Flood	0.9983	0.9990	0.9986	11,522
Accuracy			0.9332	69,134
Macro Avg	0.9486	0.9332	0.9913	69,134
Weighted Avg	0.9486	0.9332	0.9913	69,134

Fig. 14 (accuracy heatmap) shows FL achieving 96.82 % on NSL-KDD (row 7, column 4) versus our 97.29 % (row 7, column 1), but **Fig. 15** reveals FL's 99.30 % precision advantage (row 7, column 4) is offset by synchronization-induced recall variability across nodes. Our centralized approach eliminates node heterogeneity—the 459,044:3016 Normal-to-Recon ratio is consistent across all training batches, whereas FL nodes with skewed local data distributions degrade global model performance despite privacy preservation benefits.

Edge deployment demonstrates promise with clear trade-offs. On WSN-DS, Edge AI achieves excellent recall (99.83 %) but reduced precision, leading to higher false positive rates. The DNN model, while showing lower recall on this dataset (90.71 %), maintains superior precision (99.71 %) and better overall balance (87.79 % F1), making it more suitable for resource-constrained environments where false alarms are costly. Blockchain-based approaches show mixed results with inconsistent precision-recall performance. On ToN-IoT, Blockchain achieves 99.24 % precision but lacks comprehensive metric reporting, hindering direct comparison. Our centralized approach provides complete metric transparency and consistent performance.

The comprehensive metric analysis demonstrates superior balance for real-world IoT deployments. A practical example illustrates this impact: in a mid-size hydroponic greenhouse of 220 m² requiring 2.7–3.6 m³ irrigation at summer peak demand, false positives trigger unnecessary alerts and manual interventions, while false negatives allow attacks to proceed undetected [59]. High precision (99.15 % on UNSW-NB15) minimizes unnecessary alerts, reducing operational overhead. Balanced recall (83.94 %) ensures most attacks are detected, preventing resource waste. If a spoofed soil-moisture sensor or DoS attack forces the pump to run one extra hour (25 % overhead), approximately 0.7–0.9 m³ of water and 0.4–0.7 kWh of electricity are wasted. The current EU grid factor (~0.24 kg CO₂ kWh⁻¹) means that single incident adds 0.1–0.2 kg CO₂-eq. The balanced detection approach optimizes both attack prevention and false alarm reduction [60].

The variations in performance observed across different studies in our comparative analysis could stem from several factors, including variations in data preprocessing pipelines, evaluation protocols, and metric reporting standards. Models like CNN+LSTM or Transformers showing high accuracy but poor recall suggest potential evaluation inconsistencies or dataset-specific overfitting. Standardized preprocessing and comprehensive metric reporting provide transparent performance assessment, though exact

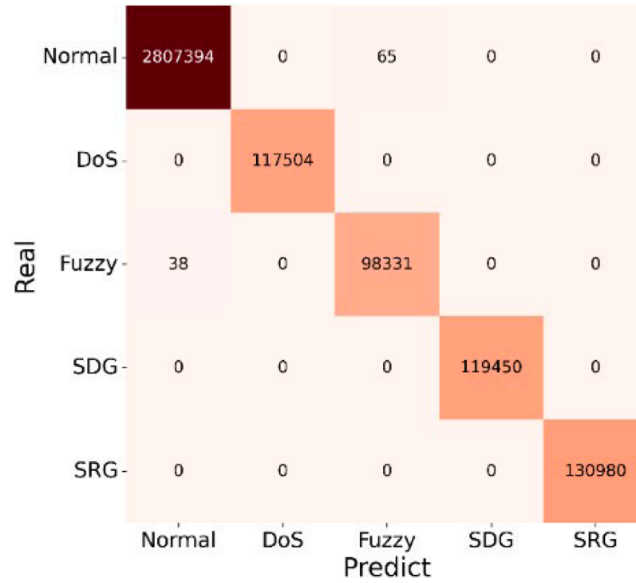


Fig. 8. Confusion matrix for the proposed multi-class IDS on the Car-Hacking dataset, separating Normal traffic from DoS, Fuzzy, Scan-and-Gathering (SDG), and Social Relationship Graph (SRG) attacks.

Table V

Classification report on Car-Hacking. Performance is near-perfect across Normal, DoS, Fuzzy, SDG, and SRG, reflecting very high separability and minimal residual errors.

Class	Precision	Recall	F1-Score	Support
Normal	1.0000	1.0000	1.0000	2807,459
DoS	1.0000	1.0000	1.0000	117,504
Fuzzy	0.9993	0.9996	0.9995	98,369
SDG	1.0000	1.0000	1.0000	119,450
SRG	1.0000	1.0000	1.0000	130,980
Accuracy			1.0000	3273,762
Macro Avg	0.999996	0.999996	0.999996	3273,762
Weighted Avg	1.0000	1.0000	1.0000	3273,762

replication remains challenging due to potential discrepancies in library versions, hardware configurations, and occasional unavailability of original implementations.

The analysis suggests several improvement directions: targeted recall enhancement for specific dataset characteristics, development of adaptive architectures that adjust precision-recall trade-offs based on deployment context, and integration of the lightweight DNN with complementary techniques to address specific weaknesses while maintaining overall efficiency.

5.1. Limitations and challenges

Although the benchmark demonstrates strong accuracy and reproducible pipeline, several important caveats remain. All evaluations rely on static, publicly available datasets, each with capture context, labeling conventions, and inherent biases. Despite applying SMOTE and Tomek Links to rebalance classes, production IoT traffic often exhibits dynamic and evolving attack patterns that no fixed preprocessing can perfectly anticipate. Models validated offline may suffer degraded performance when confronted with novel or heavily skewed traffic distributions in the field.

Experiments were conducted entirely offline on historical traces using desktop configuration (Python 3.8.4, TensorFlow 2.2, Keras 2.4.0 on Intel i7 14th-Gen with 16 GB RAM and GTX 4060 Ti). While this setup guarantees reproducibility, it does not reflect constraints of many real-world IoT deployments. Translating the DNN to resource-constrained devices requires quantization, pruning, or conversion to frameworks like TensorFlow Lite—or native C implementations—each potentially introducing accuracy degradation, numerical instability, or integration challenges. Active work continues on these conversions to evaluate impact on model fidelity and runtime efficiency.

Moving from offline analysis to live, streaming inference poses non-trivial hurdles. Production environments require continuous data ingestion with low-latency preprocessing, robust memory management, and resilience to concept drift as traffic and attack methods evolve. Measuring true end-to-end performance—throughput, latency, energy consumption, and fail-safe behavior—requires field trials on representative hardware, which are planned but beyond the current study's scope [61].

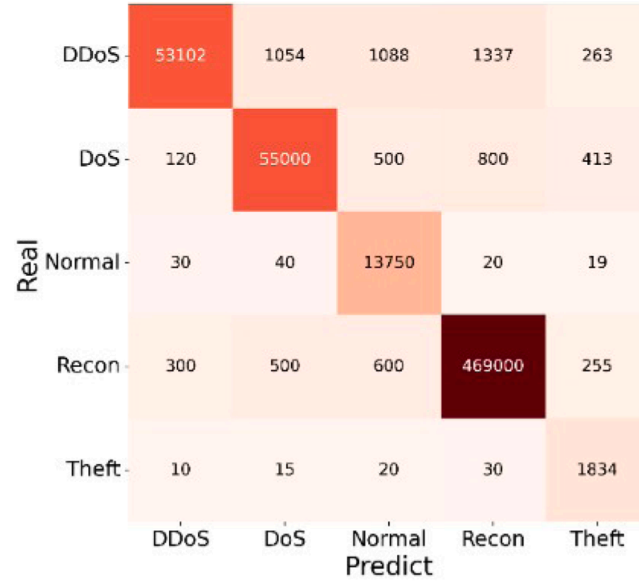


Fig. 9. Confusion matrix for the proposed multi-class IDS on the NF-BoT-IoT-v2 dataset, distinguishing DDoS, DoS, Normal, Recon, and Theft traffic.

Table VI

Classification report on NF-BoT-IoT-v2. The IDS excels on DDoS, DoS, and Recon, while Theft and Normal show comparatively lower scores—consistent with class imbalance and harder boundary cases.

Class	Precision	Recall	F1-Score	Support
DDoS	0.990	0.934	0.961	56,844
DoS	0.972	0.968	0.971	56,833
Normal	0.862	0.992	0.922	13,859
Recon	0.995	0.997	0.996	470,655
Theft	0.659	0.960	0.781	1909
Accuracy			0.989	600,100
Macro Avg	0.896	0.970	0.926	600,100
Weighted Avg	0.989	0.989	0.988	600,100

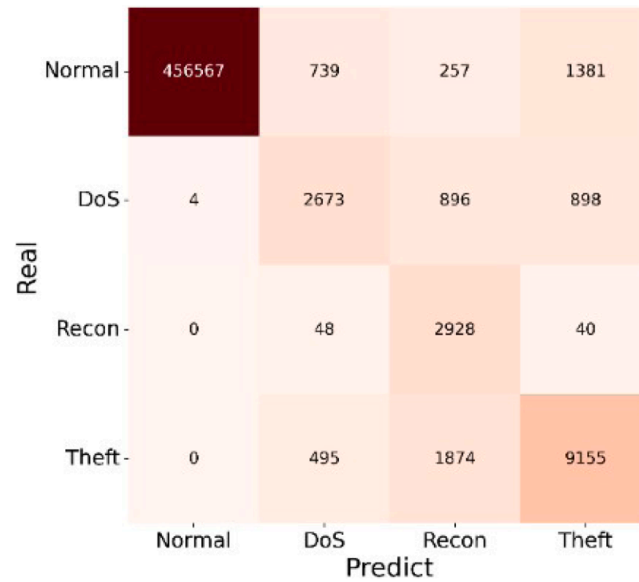


Fig. 10. Confusion matrix for the proposed multi-class IDS on the NF-UNSW-NB15-v2 dataset, distinguishing Normal, DoS, Recon, and Theft classes.

Table VII

Classification report on NF-UNSW-NB15-v2. Excellent detection for Normal; moderate for DoS; and weaker precision on Recon. Macro averages capture the challenge posed by minority classes.

Class	Precision	Recall	F1-Score	Support
Normal	1.0000	0.9946	0.9973	459,044
DoS	0.6759	0.5979	0.6345	4471
Recon	0.4836	0.9708	0.6456	3016
Theft	0.7979	0.7944	0.7962	11,524
Accuracy			0.9859	478,055
Macro Avg	0.7393	0.8394	0.7684	478,055
Weighted Avg	0.9888	0.9859	0.9868	478,055

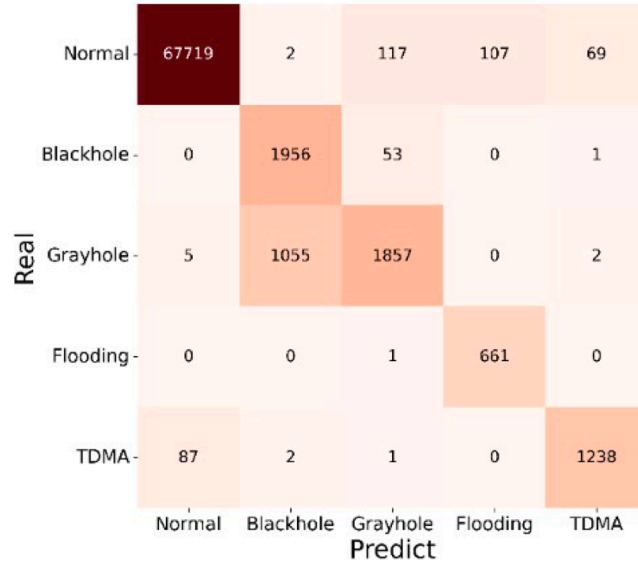


Fig. 11. Confusion matrix for the proposed multi-class IDS on the WSN-DS dataset, distinguishing Normal traffic from Blackhole, Grayhole, Flooding, and TDMA attacks.

Table VIII

Classification report on WSN-DS. Very high scores for Normal, Flooding, and TDMA; Blackhole and Grayhole remain the most challenging classes, slightly lowering macro averages.

Class	Precision	Recall	F1-Score	Support
Normal	0.9986	0.9957	0.9972	68,014
Blackhole	0.6488	0.9731	0.7785	2010
Grayhole	0.9152	0.6362	0.7506	2919
Flooding	0.8607	0.9985	0.9245	662
TDMA	0.9450	0.9322	0.9386	1328
Accuracy			0.9800	74,933
Macro Avg	0.8737	0.9071	0.8779	74,933
Weighted Avg	0.9838	0.9800	0.9800	74,933

In parallel, we recognize that evaluating true out-of-domain robustness is essential for deployment in diverse agricultural IoT environments. To this end, we will conduct cross-dataset generalization experiments by training our DNN on one public benchmark (e. g., BoT-IoT) and evaluating it directly on another (e.g., CICIDS2017) without retraining. We will report the resulting drops in accuracy, precision, and F1-score to quantify how performance degrades when exposed to completely unseen traffic distributions, thereby informing necessary adaptation strategies or retraining schedules.

While the single DNN achieves $\geq 97\%$ accuracy across eighteen datasets, no single architecture can optimally address every IoT scenario. Ensemble strategies or expert-in-the-loop systems remain unexplored, where a lightweight detector might trigger deeper analyses by more complex models. Designing and evaluating such hybrid pipelines—through expert voting or stacked inference—to balance detection speed, resource use, and adaptability remains an open challenge under investigation.

Reliance on publicly documented methodological details means that library version mismatches, missing hyperparameter settings, and unreported data splits occasionally forced reasonable assumptions. Such ambiguities can subtly affect reproducibility and

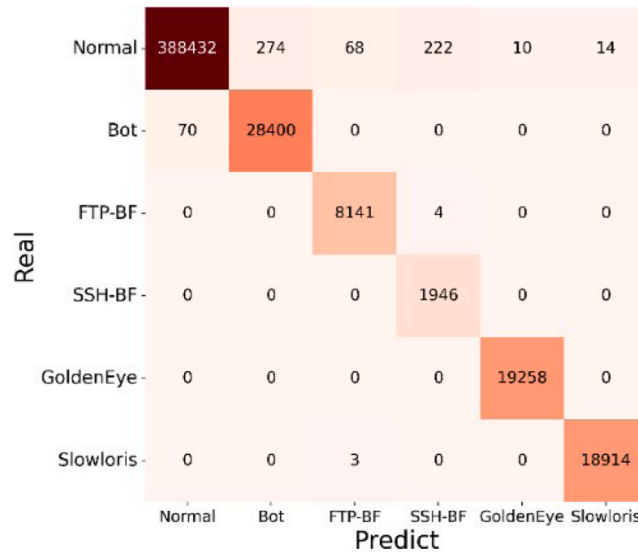


Fig. 12. Confusion matrix for the proposed multi-class IDS on the CICIDS2018 dataset, distinguishing Normal, Bot, FTP Brute-Force (FTP-BF), SSH Brute-Force (SSH-BF), GoldenEye, and Slowloris attacks.

Table IX

Classification report on CICIDS2018. Near-ceiling performance overall; SSH-BF shows the most noticeable drop in precision, while other attack types remain highly separable.

Class	Precision	Recall	F1-Score	Support
Normal	0.9998	0.9985	0.9992	389,020
Bot	0.9904	0.9975	0.9940	28,470
FTP-BF	0.9914	0.9995	0.9954	8145
SSH-BF	0.8959	1.0000	0.9451	1946
GoldenEye	0.9999	1.0000	0.9999	19,258
Slowloris	1.0000	0.9998	0.9999	18,917
Accuracy			0.9999	446,499
Macro Avg	0.9947	0.9992	0.9993	446,499
Weighted Avg	0.9993	0.9988	0.9990	446,499

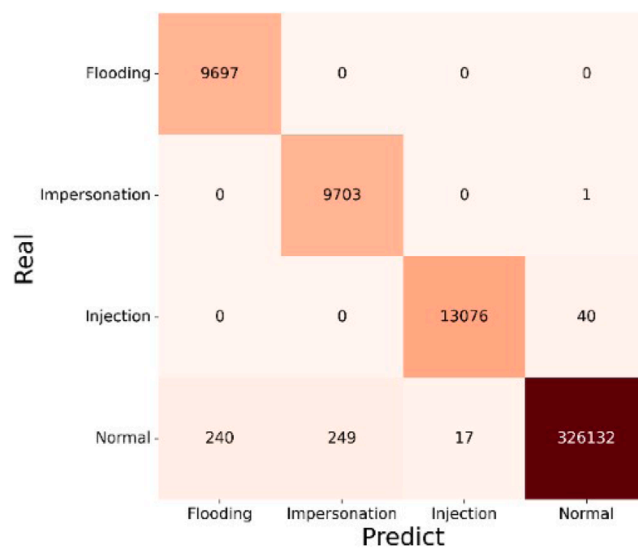


Fig. 13. Confusion matrix for the proposed multi-class IDS on the AWID dataset, distinguishing Flooding, Impersonation, Injection, and Normal classes.

Table X

Classification report on AWID. Consistently strong results across Flooding, Impersonation, Injection, and Normal, yielding high macro averages and overall accuracy.

Class	Precision	Recall	F1-Score	Support
Flooding	0.9758	1.0000	0.9878	9697
Impersonation	0.9750	0.9999	0.9873	9704
Injection	0.9987	1.0000	0.9994	13,076
Normal	1.0000	0.9985	0.9992	326,638
Accuracy			0.9986	359,115
Macro Avg	0.9874	0.9996	0.9934	359,115

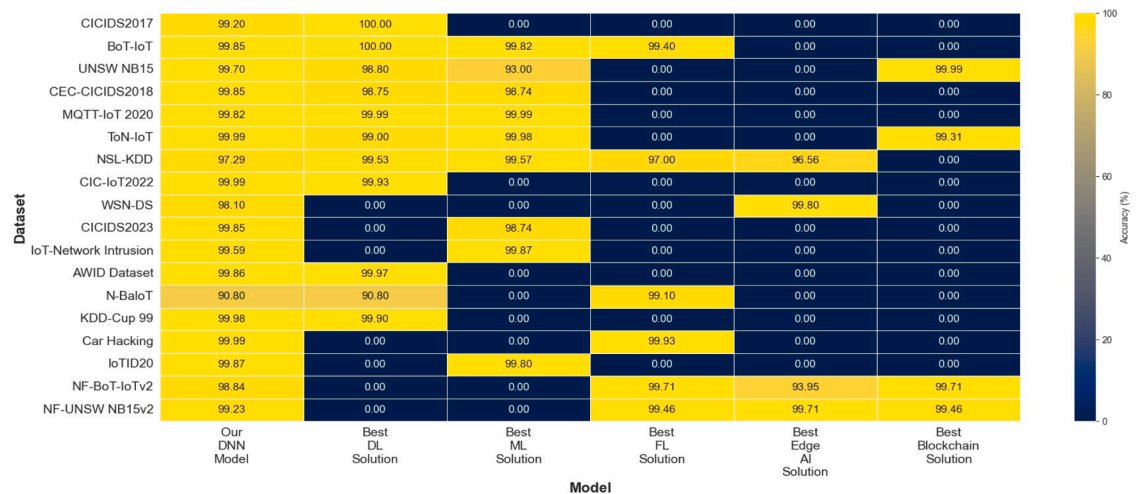


Fig. 14. Heat-map of accuracy (%) across datasets (rows) and model families (columns: Our DNN Model, Best DL Solution, Best ML Solution, Best FL Solution, Best Edge-AI Solution, Best Blockchain Solution). Cell values are percentages; yellow = higher, blue = lower. 0.00 marks combinations not evaluated or not reported in the literature.

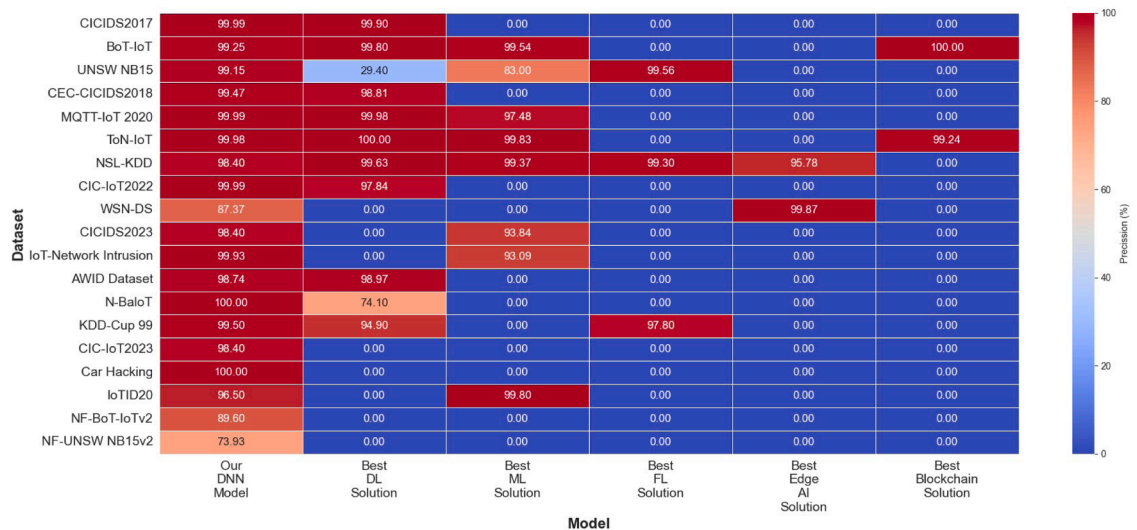


Fig. 15. Heat-map of precision (%) across datasets (rows) and model families (columns as in Fig. 14). Precision measures the proportion of predicted attacks that are correct; red = higher, blue = lower. 0.00 indicates non-evaluated / non-reported combinations.

comparability. Mitigation includes providing containerized environments and full code listings, but external studies lacking similar transparency may yield inconsistent benchmarks.

Finally, broader concerns around privacy, security, and maintainability warrant attention. Deploying IDS models close to sensitive

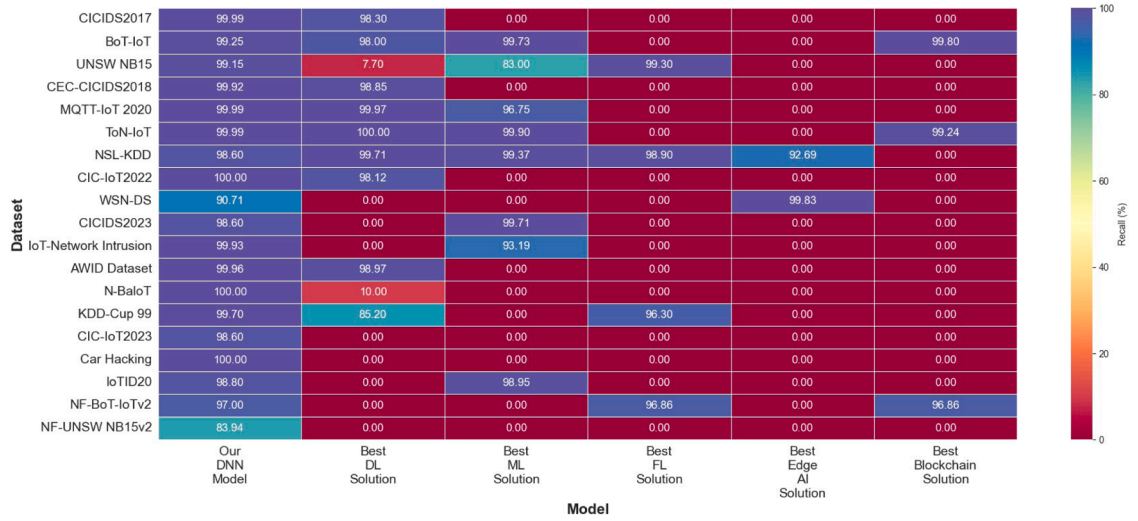


Fig. 16. Heat-map of recall (%) across datasets (rows) and model families (columns as in Fig. 14). Recall measures the proportion of actual attacks detected; blue = higher, red = lower. 0.00 denotes non-evaluated / non-reported combinations.

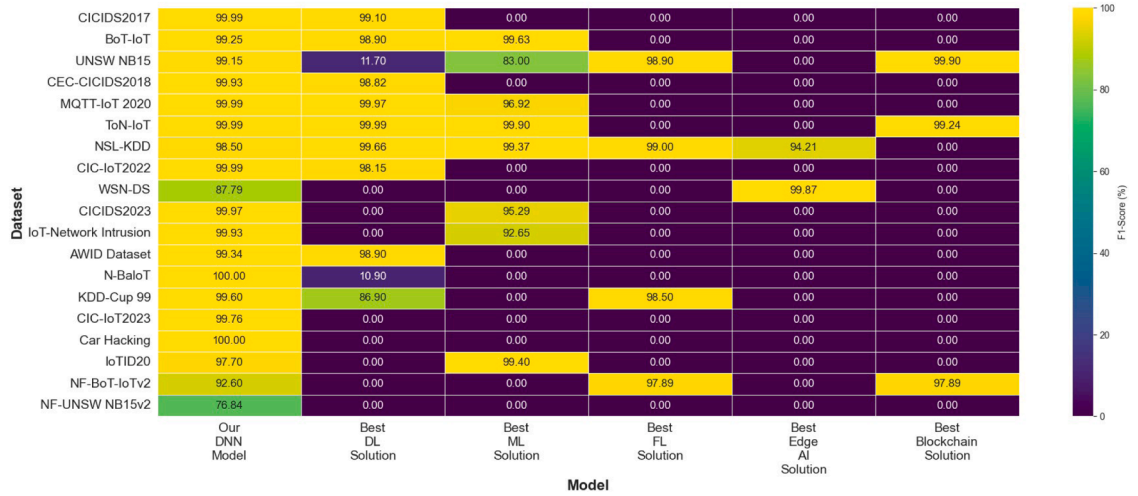


Fig. 17. Heat-map of F1-score (%) (harmonic mean of precision and recall) across datasets (rows) and model families (columns as in Fig. 14). Yellow = higher, purple = lower. 0.00 marks non-evaluated / non-reported combinations.

data streams raises questions about model tampering, data leakage during preprocessing, and long-term maintenance as IoT ecosystems scale. Future work must address secure model updates, encrypted inference, and interoperable standards to ensure that high accuracy does not compromise system integrity or user trust.

6. Conclusion

The comparative analysis demonstrates that the proposed lightweight DNN consistently delivers high-fidelity intrusion detection across heterogeneous IoT datasets: accuracy remains above 97 % and averages 99.14 %, surpassing the best-reported DL, ML, FL, Edge AI and Blockchain-enabled baselines. Because feature learning is entirely data-driven, generalization is achieved without manual engineering, and performance holds even where distributed or decentralized schemes suffer from synchronization latency or device heterogeneity. Preliminary results on the Farm-Flow greenhouse trace corroborate this performance in a real smart-agriculture setting, underscoring the approach's relevance to secure and sustainable agri-food IoT. Future work includes compressing the DNN for real-time microcontroller use, trialing a centralized-plus-federated/blockchain update scheme, and integrating unsupervised, semi-supervised and explainable modules for novel-attack detection. Current results are constrained by the fidelity of publicly available reference implementations; standardized, open-source benchmarks, adaptive or online training pipelines and lightweight hardware accelerators will be essential to close the remaining gaps between laboratory-grade accuracy and field-deployable resilience.

Beyond smart agriculture, the lightweight DNN is equally well suited to a variety of real-world IoT scenarios. In smart homes, the model can run on edge routers to detect attacks on cameras, thermostats or voice assistants while minimizing false alarms. In connected healthcare, continuous monitoring of medical devices (e.g., pulse oximeters, insulin pumps) can guard against malicious manipulation. In Industry 4.0, sensor networks in automated production lines benefit from low-latency, bandwidth-efficient detection. In smart cities, infrastructure such as traffic lights, street lighting and EV charging stations can be secured without compromising energy efficiency. Finally, in distributed energy systems, embedding the model in lightweight SCADA gateways enables real-time intrusion detection and grid stability. Thanks to its compact size (~1.2 M parameters), low inference cost and reproducible pre-processing pipeline, deployment on edge gateways and even microcontrollers becomes feasible, supporting scalable, low-power security across diverse IoT environments.

CRediT authorship contribution statement

A. Villafranca: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Maria-Dolores Cano:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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